

Complete MERN Stack Deployment Guide (AWS EC2 + Local MongoDB)

This guide deploys a MERN stack application on a fresh Ubuntu 22.04 AWS EC2 instance using: React (Vite), Express, Node.js, MongoDB (Local), PM2, Nginx

1. Launch EC2 Instance

Recommended Settings

- Ubuntu Server 22.04
- t2.micro or t3.micro
- 20 GB storage recommended

Security Group Ports

Allow:

- 22 → SSH
 - 80 → HTTP
 - 443 → HTTPS
 - 5000 → Optional backend testing
-

2. Connect To EC2

Run on your local machine:

```
chmod 400 your-key.pem
```

```
ssh -i your-key.pem ubuntu@YOUR_EC2_PUBLIC_IP
```

3. Update Server

```
sudo apt update
```

```
sudo apt upgrade -y
```

4. Install Node.js 20

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
```

```
sudo apt install nodejs -y
```

Verify:

node -v

npm -v

5. Install Git and Nginx

sudo apt install git nginx -y

Enable and start nginx:

sudo systemctl enable nginx

sudo systemctl start nginx

6. Install MongoDB 7

Import MongoDB Key

curl -fsSL https://pgp.mongodb.com/server-7.0.asc | \

sudo gpg -o /usr/share/keyrings/mongodb-server-7.0.gpg \

--dearmor

Add MongoDB Repository

echo "deb [arch=amd64 signed-by=/usr/share/keyrings/mongodb-server-7.0.gpg]
https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/7.0 multiverse" | sudo tee
/etc/apt/sources.list.d/mongodb-org-7.0.list

Update Packages

sudo apt update

Install MongoDB

sudo apt install mongodb-org -y

7. Start MongoDB

sudo systemctl start mongod

```
sudo systemctl enable mongod
```

Check status:

```
sudo systemctl status mongod
```

Expected:

active (running)

8. Install PM2

```
sudo npm install -g pm2
```

9. Clone Your Project

```
git clone YOUR_GITHUB_REPO_URL
```

Example:

```
git clone https://github.com/username/project.git
```

Go inside project:

```
cd PROJECT_NAME
```

10. Setup Backend

Go to server folder:

```
cd server
```

Install backend dependencies:

```
npm install
```

11. Create Backend Environment File

If .env.example exists:

```
cp .env.example .env
```

Edit file:

```
nano .env
```

Add:

PORT=5000

NODE_ENV=production

MONGO_URI=mongodb://127.0.0.1:27017/eventdb

Save and exit:

- CTRL + O
- ENTER
- CTRL + X

12. Test Backend

node index.js

Expected:

Mongo Connected

Server running on port 5000

Stop server:

CTRL + C

13. Setup Frontend

Go to frontend:

cd ../client

Install frontend dependencies:

npm install

Build production frontend:

npm run build

This creates:

client/dist

14. Copy Frontend Build To Server

Go to root project folder:

```
cd ..
```

Copy frontend build:

```
cp -r client/dist server/client-dist
```

15. Start Backend Using PM2

Go to server:

```
cd server
```

Start app:

```
pm2 start index.js --name event-app
```

Check status:

```
pm2 status
```

Save PM2 process:

```
pm2 save
```

Enable PM2 auto-start:

```
pm2 startup
```

Copy-paste the command PM2 provides.

16. Configure Nginx

Edit nginx config:

```
sudo nano /etc/nginx/sites-available/default
```

Replace everything with:

```
server{
```

```
    listen 80;
```

```
    server_name _;
```

```
    location / {
```

```
        proxy_pass http://localhost:5000;
```

```
proxy_http_version 1.1;
```

```
proxy_set_header Upgrade $http_upgrade;
```

```
proxy_set_header Connection 'upgrade';
```

```
proxy_set_header Host $host;
```

```
proxy_cache_bypass $http_upgrade;
```

```
}
```

```
}
```

Save and exit:

- CTRL + O
 - ENTER
 - CTRL + X
-

17. Test Nginx Configuration

```
sudo nginx -t
```

Expected:

```
syntax is ok
```

```
test is successful
```

18. Restart Nginx

```
sudo systemctl restart nginx
```

19. Open Website

Visit:

```
http://YOUR_EC2_PUBLIC_IP
```

Your MERN application should now work.
